

Position:

$$x = x_s + r \cos \theta$$

$$y = y_s + r \sin \theta$$

$$\dot{x} = \dot{x}_s - r \sin \theta \dot{\theta}$$

$$\dot{y} = \dot{y}_s + r \cos \theta \dot{\theta}$$

$$V = \dot{x} + \dot{y}$$

Now, for Kinetic energy

$$\frac{1}{2} m \dot{V}^2 = K.E \Rightarrow \frac{1}{2} m (\dot{x} + \dot{y})^2$$

$$\Rightarrow \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + 2 \dot{x} \dot{y})$$

$$\Rightarrow \frac{1}{2} m (\dot{x}_s^2 + r^2 \dot{\theta}^2 + \dot{y}_s^2 + r^2 \cos^2 \theta \dot{\theta}^2 + 2(\dot{x}_s - r \sin \theta \dot{\theta})(\dot{y}_s + r \cos \theta \dot{\theta}))$$

$$\Rightarrow \frac{1}{2} m (\dot{x}_s^2 + r^2 \dot{\theta}^2 (\sin^2 \theta + \cos^2 \theta) + \dot{y}_s^2 + 2(\dot{x}_s - r \sin \theta \dot{\theta})(\dot{y}_s + r \cos \theta \dot{\theta}))$$

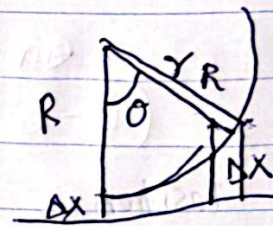
$$\Rightarrow \frac{1}{2} m \dot{x}_s^2 + \frac{1}{2} m r^2 \dot{\theta}^2 + \frac{1}{2} m \dot{y}_s^2 - \text{Kinetic energy of mass } T_{\text{mass}}$$

K.E //

$$\Rightarrow \frac{1}{2} m \dot{x}_s^2 + \frac{1}{2} m r^2 \dot{\theta}^2 + \frac{1}{2} m \dot{y}_s^2 + m (\dot{x}_s - r \sin \theta \dot{\theta})(\dot{y}_s + r \cos \theta \dot{\theta}) //$$

For potential energy:

$$\begin{aligned}
 U &= mgh \\
 &= mg(R - r \cos \theta) \\
 &= -mgr \cos \theta \quad (\text{dropped Const}) \quad \text{at } t=0 \text{ is } (R - r)
 \end{aligned}$$



K.E of Tshell:

$$x_c = R \dot{\phi}, \quad y_c = 0$$

$$V = R \dot{\phi} = \frac{1}{2} m v_{cm}^2 = \frac{1}{2} M R^2 \dot{\phi}^2 \rightarrow \text{translation}$$

$$T_{\text{shell rotation}} = \frac{1}{2} I_s \dot{\phi}^2$$

$$\begin{aligned}
 T_{\text{shell}} &= \frac{1}{2} M R^2 \dot{\phi}^2 + \frac{1}{2} I_s \dot{\phi}^2 \\
 &= \frac{1}{2} (M R^2 + I_s) \dot{\phi}^2
 \end{aligned}$$

$$T = T_{\text{shell}} + T_{\text{mass}}$$

Damping $m_1 R^2 \dot{\theta}^2 + m_2 R^2 \dot{\theta}^2 + \dot{\phi}^2 (2I_1 + 2I_2 + m_1 R^2 + m_2 R^2)$
 $(\dot{\phi} - \dot{\theta})^2 r d + \dot{\phi}^2 q d =$

we use Rayleigh dissipation function to describe the rolling resistance on shell and bearing b/w the mass & track and shell.

$$\mathcal{F} = \frac{1}{2} b_{\phi} \dot{\phi}^2 + \frac{1}{2} b_{\psi} (\dot{\theta} - \dot{\phi})^2$$

Using Euler Lagrange eq.:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = - \frac{\partial \mathcal{F}}{\partial \dot{\phi}}$$

$$L = T - V = \frac{1}{2} [(m + m) R^2 + I_s] \dot{\phi}^2 + m R r \dot{\phi} \dot{\theta} \cos \theta + \frac{1}{2} m r^2 \dot{\theta}^2 + m g r \cos \theta.$$

$$\frac{\partial L}{\partial \dot{\phi}} = [(m + m) R^2 + I_s] \dot{\phi} + m R r \dot{\theta} \cos \theta.$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = [(m + m) R^2 + I_s] \ddot{\phi} + m R r \ddot{\theta} \cos \theta - m R r \dot{\theta}^2 \sin \theta$$

$$\frac{\partial \mathcal{F}}{\partial \dot{\phi}} \Rightarrow Q_{\phi} = -b_{\phi} \dot{\phi} + b_{\psi} (\dot{\theta} - \dot{\phi})$$

$$[(M+m)R^2 + I_S] \dot{\phi} + m R r \dot{\theta} \cos \theta - m R r \dot{\theta}^2 \sin \theta = -b_{\phi} \dot{\phi} + b_{\psi} (\dot{\theta} - \dot{\phi})$$

$$\therefore T_{\text{mag}} = m R r \cos \theta \ddot{\phi} + m r^2 \ddot{\theta} + m g r \sin \theta + b_{\psi} (\dot{\theta} - \dot{\phi}) //$$

$$T_{\phi} = \frac{\partial L}{\partial \dot{\phi}} = \frac{\partial}{\partial \dot{\phi}} \left[\frac{1}{2} (M+m) R^2 \dot{\phi}^2 + m R r \dot{\theta} \dot{\phi} \cos \theta - \frac{1}{2} I_S \dot{\phi}^2 - \frac{1}{2} I_S \dot{\theta}^2 - m g r \cos \theta \right]$$

$$T_{\phi} = (M+m) R^2 \dot{\phi} + m R r \dot{\theta} \cos \theta$$

$$T_{\theta} = \frac{\partial L}{\partial \dot{\theta}} = \frac{\partial}{\partial \dot{\theta}} \left[\frac{1}{2} (M+m) R^2 \dot{\phi}^2 + m R r \dot{\theta} \dot{\phi} \cos \theta - \frac{1}{2} I_S \dot{\phi}^2 - \frac{1}{2} I_S \dot{\theta}^2 - m g r \cos \theta \right]$$

$$T_{\theta} = m R r \dot{\phi} \cos \theta - I_S \dot{\theta}$$

$$T_{\phi} = (M+m) R^2 \dot{\phi} + m R r \dot{\theta} \cos \theta$$